

ML Problem Solving

Data as Vectors, Regression vs Classification

Consider the following table, representing information about a handful of people.

Name	Gender	Degree	Postcode	Age	Annual salary
Aditya	M	MSc	W21BG	36	89563
Bob	M	PhD	EC1a1BA	47	123543
Chloe	F	BEcon	SW1a1BH	26	23989
Daisuke	M	BSc	SE207AT	68	138769
Elisabeth	F	MBA	SE10AA	33	113888

1. Create two versions of design matrices comprised of only numerical values to represent the data provided. Each version should encode the information provided differently, e.g. the MSc degree value should be different in the two design matrices. The difference needs to be present in at least one column.

2. Draw the data points from your design matrices in a coordinate systems where the x-axis is the age and the y-axis is the salary. Draw a linear model on your coordinate system that graphically has the smallest possible prediction error.

3. Draw the data points from your design matrix in a coordinate system where the x-axis is the degree and the y-axis is the salary. Draw a linear model on your coordinate system that graphically has the smallest possible prediction error.

4. Propose two loss functions that will allow to calculate the error between the actual value present in the linear model and the predicted value. Calculate the loss values for your 2 design matrices and your 2 features (age, degree). The loss should be the sum of the loss of each datapoints present. A total of 8 loss values are expected (varying encoding, feature and loss function).

5. For the following tasks, which are classification and which are regression ones?

- (____) Ordering a sandwich at Ctrl-Alt-Deli
- (____) Estimating the size of a building from images
- (____) Predicting stock prices
- (____) Choosing stocks to purchase
- (____) Neural network outputs
- (____) Making a move in checkers
- (____) Ranking a user's Netflix shows
- (____) Selecting from different plans for moving a robot
- (____) Choosing a product from an Amazon search results page